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Formation of polychlorinated dibenzo-*p*-dioxins and dibenzofurans (PCDD/F) in oxidation of captan pesticide ☆

Kai Chen [a], Dominika Wojtalewicz [a], Mohammednoor Altarawneh [b], John C. Mackie [a,c], Eric M. Kennedy [a], Bogdan Z. Dlugogorski [a,*]

[a] *Process Safety and Environmental Protection Group, School of Engineering, The University of Newcastle, Callaghan, NSW 2308, Australia*
[b] *Chemical Engineering Department, Al-Hussein Bin Talal University, Ma'an, Jordan*
[c] *School of Chemistry, The University of Sydney, NSW 2006, Australia*

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Abstract

This study assessed the emission of polychlorinated dibenzo-*p*-dioxins and polychlorinated dibenzofurans (PCDD and PCDF) from captan, a commonly used fungicide, in vapour-phase oxidative pyrolysis under conditions similar to those encountered in fires and burning biomass contaminated or treated with pesticides. The laboratory-scale apparatus consisted of a pesticide vaporiser, a tubular reactor and a product sampling system. The sampling train comprised tandem XAD-2 resin cartridges to trap PCDD/F as well as an activated charcoal tube to capture the organic volatile compounds (VOC). The analyses of PCDD/F were conducted by means of high resolution gas chromatograph (HRGC) – ion trap mass spectrometer (ITMS) and analyses of VOC by HRGC – quadrupole mass spectrometer (QMS). Substantially more PCDF formed than PCDD in the oxidative pyrolysis of captan, with higher yield of total PCDD/F observed at longer residence time. As indicated by the homologue distribution of PCDD/F, only mono to tetra chlorinated congeners were detected in our measurements, with 4-monochlorinated dibenzofuran (4-MCDF) ranking as the most abundant congener. The results of VOC analysis revealed benzene and chlorinated benzenes as important PCDD/F precursors. Combining the experimental measurements and the results of quantum chemical calculations, we established the reaction pathways for formation of PCDD/F from the vapour-phase oxidative pyrolysis of this widely employed fungicide.
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Keywords: Oxidative pyrolysis; PCDD/F; Dioxins; Precursors; Reaction pathways

1. Introduction

Belonging to the sulfenimide group of fungicides, captan is widely applied to vegetable crops, fruits and ornamentals [1,2]. First registered for fruit tree use in 1949, it ranks as the second most abundantly employed fungicide on peach and

☆ This manuscript presents a detailed experimental investigation to assess the propensity of captan, a widely used fungicide, to form polychlorinated dibenzo-*p*-dioxins and polychlorinated dibenzofurans (PCDD/F) under oxidative pyrolysis conditions.

* Corresponding author. Fax: +61 2 4921 6893.
E-mail address: Bogdan.Dlugogorski@newcastle.edu.au (B.Z. Dlugogorski).

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apple orchards in the USA [3]. As the active ingredient for more than 160 products, the statistics showed that 215,301 acres in California (2007) were treated with captan [4]. Furthermore, captan has broad industrial applications, especially in cosmetics, paints and textiles, with around 58.8–163.9 tonnes consumed each year in these applications [5,6].

Owing to its high chlorine content (35.4%) and large annual production, captan was included in European TOXFIRE project, and found to form PCDD/F; in spite of no apparent PCDD/F precursor moieties in its structure (refer to Fig. 3). The project evaluated the emission factors of major pollutants, including PCDD/F from chemical fires [7,8]. Moreover, captan residue left on processed tobacco may produce PCDD/F during the burning of cigarettes [9]. PCDD/F denoted as dioxins in chemist's vernacular, are classified as highly carcinogenic and persistent organic pollutants and their emissions have attracted significant research and regulatory scrutiny. The domestic combustion of contaminated biomass and accidental chemical fires, involving conditions with propensity for PCDD/F formation, has been identified as significant sources of PCDD/F [10]. In these combustion scenarios, the presence of captan will likely elevate the emissions of PCDD/F.

Unfortunately, the present understanding of mechanisms of PCDD/F formation in the combustion of pesticides is quite limited, compared to those of chlorobenzene, chlorophenol and PVC [7,11–13]. Our group has reported the emission of PCDD/F from the oxidative pyrolysis of permethrin and tebuconazole (wood preservatives) [14,15]. These chemicals include chlorine and incorporate phenyl, biphenyl and diphenyl ether precursors to PCDD/F in their structures. While the thermal decomposition of captan has been studied, only a few publications have examined the emission of PCDD/F. One study has reported the PCDD/F emission factors from solid captan heated in a DIN furnace [7]. With only the 17 regulated congeners covered in these measurements, they presented neither a description of precursors nor a detailed mechanistic study of the PCDD/F formation.

This article reports the results of the oxidative pyrolysis of gas-phase captan, elucidating the thermal pathways for the formation of PCDD/F. The aim was to investigate the potential of captan combustion as a possible source of dioxins. The experiments were performed under conditions encountered in fires. We also conducted the analyses of volatile organic compounds (VOC), attempting to identify the PCDD/F precursors. The in-house PCDD/F analysis enabled us to scrutinise a wide range of isomers, compared to PCDD/F analyses performed by commercial laboratories set up to report only 17 toxic congeners, allowing us to establish the reaction mechanism.

2. Experimental

2.1. Brief description of the apparatus

The experiments involved generating dilute stream of captan for oxidation in a tubular reactor, with the details of the apparatus (Fig. 1) described elsewhere [16]. Briefly, the vaporiser consisted of an 8 mm i.d. polytetrafluoroethylene (PTFE) tube installed horizontally in a GC oven (Shimadzu), heated at 150 °C to slowly evaporate the solid captan (sourced from TCI) under the flow of pure nitrogen (99.999%). Oxygen mixed with the vapour/N_2 stream prior to entering the reactor, resulting in an O_2 concentration of 6.0 (±0.5)% in N_2 (v/v); uniform mixing achieved by a plug of pre-cleaned glass wool. The average evaporation rate was measured as 0.042 (±10%) mg min^{-1}, corresponding to a concentration of captan of 0.18 (±0.02)% (w/w). A transfer line (175 °C), coupled the vaporiser to the tubular reactor (5 mm i.d.), and with the latter heated by a three zone furnace (Labec). Two 4 mm o.d. rods were inserted into both ends of the reactor to minimise the decomposition outside the reaction zone and to control the residence time (RT). We afforded three set values of RT (0.5, 1 and 5 s) to perform the oxidation of captan at 600 °C, temperature encountered both in the pyrolysis and cooling down zones in fires. This temperature also falls within the temperature window of the gas phase formation of PCDD/F [17]. Each experiment continued for 6 h with product gases flowing through two XAD-2 resin cartridges arranged in tandem (200 mg of resin in the front cartridge and 75 mg in the back), as well as a chilled impinger containing dichloromethane (DCM). Volatile organic compounds (VOC) were trapped on activate charcoal (approximate

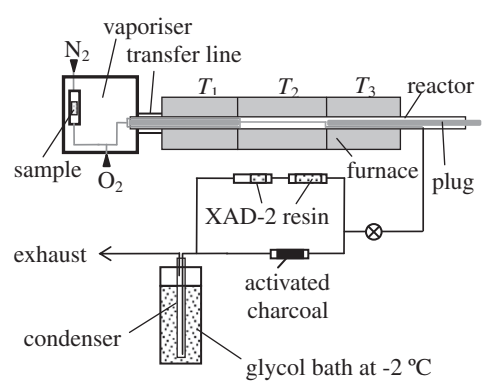

Fig. 1. A sketch of the experimental apparatus.

100 mg) in separate experiments that involved sampling for 1 h.

2.2. Sample clean-up and preparation

The chemicals adsorbed on the XAD-2 resin were extracted with toluene for 5 h in an automated Soxhlet unit (Buchi). To collect the chemicals condensed on the reactor's walls, we filled the reactor's tube with DCM and extracted it in a shaker. The short PTFE tube coupling the reactor to the resin cartridges was also extracted with DCM in an ultra-sonic bath. After repeated extractions, we collected 50 and 20 mL solutions from washing the reactor and the PTFE tube, respectively. The extracts of XAD-2, reactor and coupling tube were concentrated separately, and then cleaned up with glass chromatography columns loaded with acid and basic alumina [15]. Subsequently, the cleaned-up solutions were concentrated again and then analysed by Varian 3800 GC and Saturn 2000 ITMS. Prior to each extraction, we washed all the glassware in detergent and annealed it at 180 °C for 1 h. For the analyses of VOC, we modified NIOSH Method 1003 for use with a MS. Samples of 1 mL CS_2 extracts were filtered and then injected (split 50) into the Varian CP3800 GC, connected to the Varian 1200 QMS.

2.3. PCDD/F analytical methodologies

The limit of quantification (LOQ) of PCDD/F standards analysed on our HRGC-ITMS instrument corresponds to between 0.6 and 11.7 pg μL^{-1} with values of each congener listed in Table S1 Supplementary material. The analyses of PCDD/F standards spiked into XAD-2 resin, after 5 h hot extraction and clean-up with alumina column, yielded good recoveries (52–101%; compared with a typical range of 25–164% required by EPA Method 1613), with recovery of each congener tabulated in Supplementary material. A blank test was performed by extracting unexposed XAD-2 resin, with results displaying no presence of PCDD/F (signal/noise < 3). We demonstrated a reliable trapping efficiency of our technique, as evidenced by no breakthrough of PCDD/F into the condenser. Monochlorinated DD/F (MCDD/F) were mainly trapped in the XAD-2 resins while the other congeners condensed, preferentially, both on the reactor's walls and on PTFE tubes prior to the resin trap (see Table S4 for results of detailed measurements). Repeated runs conducted at RT of 1 s indicated the same distribution of congeners in the sampling train as well as good reproducibility in the total yield of PCDD/F, as 21.3 and 19.8 $\mu g\ g^{-1}$ respectively.

Supplementary material includes additional details of the analytical methodologies and operation of the apparatus.

2.4. Quantum chemical computations

We have performed quantum chemical calculations to predict the initial decomposition of captan and subsequent reactions of the initial products of the decomposition. Because of its size, computations on captan itself and optimisation of transition states have been carried out by density functional theory at the B3LYP/6-31G(d) level of theory. However, to obtain reliable bond energies for initial fission, models for the N–S–CCl_3 moiety in captan have been used to enable computations to be made using higher level methods. Details of the quantum chemical computations are provided in Supplementary material.

3. Results and discussion

3.1. Total yield and homologue distribution of PCDD/F

Table 1 summarises the experimental measurements of the gas phase oxidative pyrolysis of captan at 600 °C. As RT increases, we observe higher yield of total PCDD/F. The oxidation favoured the gas phase formation of PCDF, with the yield of PCDD amounting to less than 10% of that of PCDF. Figure 2 presents the homologue distributions, with mono- and dichlorodibenzofurans (DCDF) displayed next to the total PCDD/F, in the top pane of the figure; note different scales used. Although conducted at different RT, the gas phase oxidation of captan rendered a similar congener pattern; viz., mono- to trichlorinated DD/F (TriCDD/F) and tetrachlorinated dibenzofuran (TCDF), with no quantifiable levels of tetrachlorinated dibenzo-*p*-dioxin (TCDD) and no higher chlorinated congeners observed in all measurements.

MCDF exhibit a significant increase in yields with the prolonged RT. The yield of DCDF shows the opposite trend, monotonically decreasing to 3080 ng g^{-1} at RT of 5 s. Except of MCDD, the remaining congeners display declining yields with RT increasing from 0.5 to 5 s. Both MCDD and DCDD exhibit maxima at RT of 1 s. Overall, these results indicate progressive dechlorination of PCDD/F with increasing residence time.

Table 1
Summary of the experimental program and measurements.

RT s	Reactant mg	PCDD/F ng	Yield[a] $\mu g\ g^{-1}$	PCDD/PCDF
0.5	14.2	230	16.2	0.046
1	15.3	326	21.3	0.077
5	15.5	333	21.5	0.053

[a] Yield reported as $\mu g\ g^{-1}$ captan.

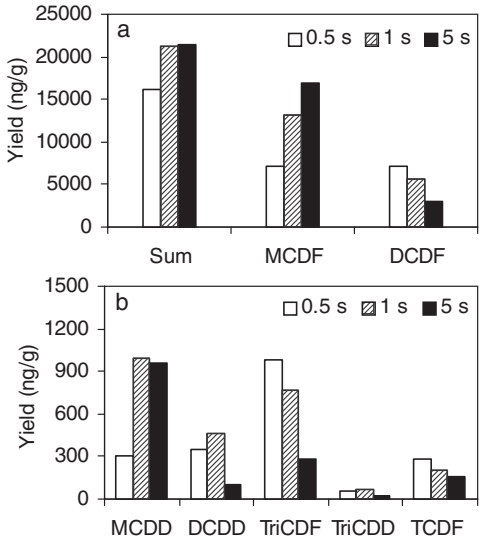

Fig. 2. Homologue profiles derived from the experimental measurements: (a) sum of PCDD/F and M/DCDF; (b) M/DCDD, TriCDD/F and TCDF.

3.2. VOC analysis and decomposition mechanism to the PCDD/F precursors

In all experiments, the GC-MS analysis of VOC identified six products: benzene, tetrachloroethylene (TCE), chlorobenzene (CB), benzonitrile (BZN), dichlorobenzene (DCB) and chlorobenzonitrile. Our interests in the present study focus on benzene, CB and DCB, which have been documented as important precursors of PCDD/F [12,17,18]. We detected all three isomers of DCB (*ortho*, *meta* and *para*), with their elution order on DB-5 column (equivalent to ours) listed in the EPA Method 8270C as *meta*-DCB < *para*-DCB < *ortho*-DCB. To confirm the retention times of important compounds, we also injected the authentic standards of PCDD/F precursors (*meta*-DCB for DCB) as well as the standards of TCE and BZN. The repeated tests of the three DCB isomers on the MS detector have demonstrated similar response factors (RR) of DCB isomers with the relative standard deviations (RSD) of less than 5%, which enabled us to quantify them with the *meta*-DCB standard. Table 2 illustrates the retention times and yields of PCDD/F precursors (mol %) in the experiments at different RT. The variation of RT has no significant influence on the yields of each precursor. Benzene is the most abundant precursor while *ortho*-DCB dominates the distribution of DCB.

The presence of these precursors indicates reactions between free Cl radicals and the aromatic ring, both formed in the captan decomposition, owing to absence of chlorine attached on the cyclohexene ring of captan. Thus, the formation of Cl radicals and aromatic rings is crucial in the production of PCDD/F. Figure 3 summarises the pathways of captan oxidation, based on the oxidation of captan at low temperatures (⩽500 °C) [19]. The experimental measurements and the quantum chemical calculations indicate that the thermal decomposition of captan, in the presence of oxygen, initiates through the fission of S—C bond, engendering CCl_3 and R1 radicals which can rapidly react with O_2. With sulfur directly oxidised to SO_2, the R1 radical can convert to *cis*-1,2,5,6-tetrahydrophthalimide (THPI). CCl_3 reacts with O_2 to produce $COCl_2$ and Cl for subsequent chlorination [20–22]. The destruction of $COCl_2$ can further release radicals including COCl and Cl [23]. In parallel, the self-combination of CCl_3 radicals forms hexachloroethane [24] and its conversion to TCE expectedly releases Cl radicals [25]. The formation of folpet and phthalimide (PI) points to the occurrence of aromatisation. Owing to strong C—H bonds off cyclohexene ring, direct bond fission is unlikely. However, the radicals, such as CCl_3 and Cl, can readily abstract hydrogen atoms from the cyclohexene ring, leading to a cyclohexenyl radical. This radical can easily lose a further H by fission with a low energy barrier, to form a cyclohexadiene ring. Eventually, further abstraction of H results in the formation of aromatic rings in both folpet and PI.

Our quantum chemical calculations indicate that H can add onto the N atom of captan or folpet, leading to the breakage of N–S bond and subsequently formation of THPI, PI and $SCCl_3$, in an exothermic process with a barrier of only 7 kcal mol^{-1} [19]. The weak C—Cl bond in $SCCl_3$ promotes the availability of Cl via bond fission to give $SCCl_2$. The further abstraction of Cl from $SCCl_2$ by H results in SCCl radical, which may lose another Cl by bond fission. The above comments indicate the availability of free Cl radicals in the oxidation of captan. BZN has been observed as a major product in a previous study of decomposition of PI [26] and benzene documented as an important product in the decomposition of BZN [27]. We proposed that the chlorination of aromatic rings (formed in the decomposition of captan) by Cl radicals serves as a main route of producing CB and DCB.

Table 2
Yields of PCDD/F precursors at different RT.

PCDD/F precursor	Retention time (min)	Yields (mole %)		
		0.5 s	1 s	5 s
Benzene	2.85	7.5	8.1	7.9
CB	6.41	0.64	0.56	0.62
meta-DCB	9.50	0.035	0.033	0.037
para-DCB	9.67	0.007	0.006	0.007
ortho-DCB	9.99	0.088	0.085	0.093

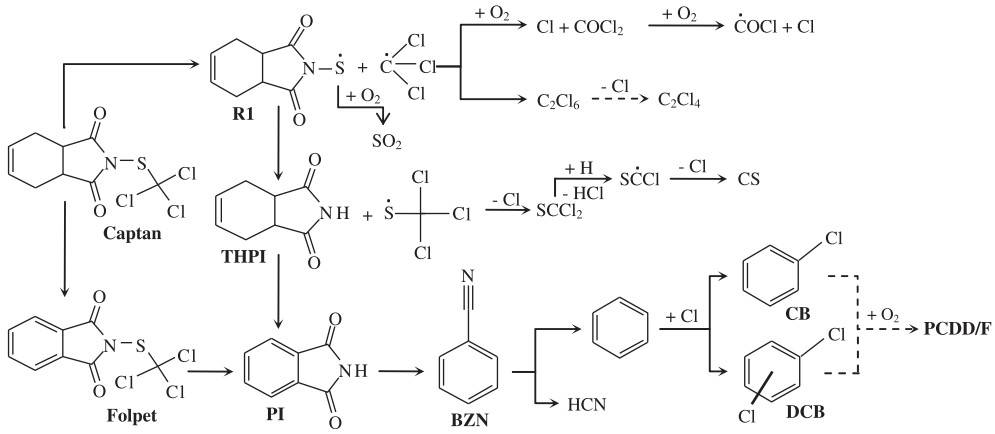

Fig. 3. Summary of pathways of captan decomposition [19].

We also attempted, but with no success, to identify phenols, another important group of precursors to PCDD/F, by attaching silica gel trap as well as bubbling the product gas into a chilled solvent trap, in spite of reports in literature that slow combustion of CB gives phenol and chlorophenols [12]. Also, one may postulate that the oxidation of captan may first generate phenols and the oxidation of phenols leads to benzene and chlorinated benzenes. However, the oxidation of chlorophenols to chlorobenzene has been reported to occur only at yield as low as 0.05% [13]. Furthermore, our quantum chemical calculations have been unable to discover pathways to phenols from either PI or THPI moieties. All these findings indicate no formation of phenols from captan in our system. Thus it is the oxidation of benzene and chlorinated benzenes that plays a dominant role in the formation of PCDD/F.

3.3. Comparison of isomers in homologue groups and the formation pathways

Figure 4 displays the isomer distribution within the three homologue groups MCDF, MCDD and DCDD, for the three residence times considered in our experiments. The identification and quantification of three homologue groups follow the injection of genuine standards (marked with asterisks in Fig. 4), and the published elution order [28]. We have observed similar distribution of MCDF, MCDD and DCDD in all experiments; viz., 4-MCDF is the most abundant isomer in MCDF homologue group while 1-MCDD dominates the distribution of MCDD. A previous study on the oxidation of chlorobenzene has revealed that decomposition initiates with an ejection of H or Cl from benzene ring [18]. Subsequently, Cl can abstract H from CB more favourably, compared to the abstraction of Cl. Oxygen attacks the available radical site resulted from H abstraction, forming three chlorophenylperoxy radicals. Those radicals can readily convert to the corresponding phenoxy radicals [12,18]. We expect that the phenoxy radicals are easily produced in our experiments, owing to the availability of Cl from the decomposition of captan.

Formation of 4-MCDF via the condensation of phenoxy radicals has been studied both experimentally and theoretically. The possible pathways involve: (1) the recombination of the carbon (hydrogen attached) centred radical with the carbon (chlorine attached) centred radical and (2) the recombination of the former chlorophenoxy radical with unchlorinated carbon-centred radical [13,29]. A carbon centred radical denotes one of

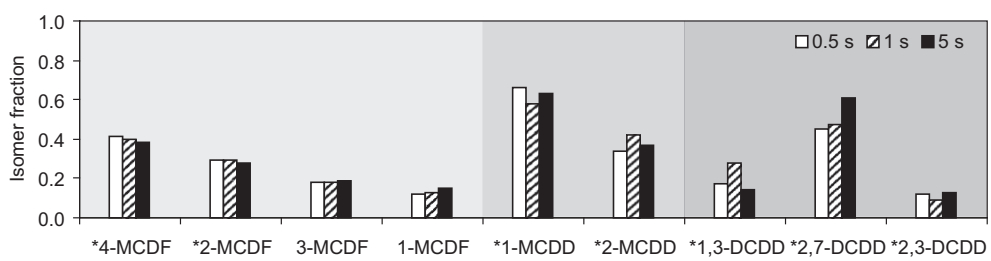

Fig. 4. Isomer patterns of MCDD/F and DCDD separated in different colour regions. ∗ confirmed by injecting genuine standard.

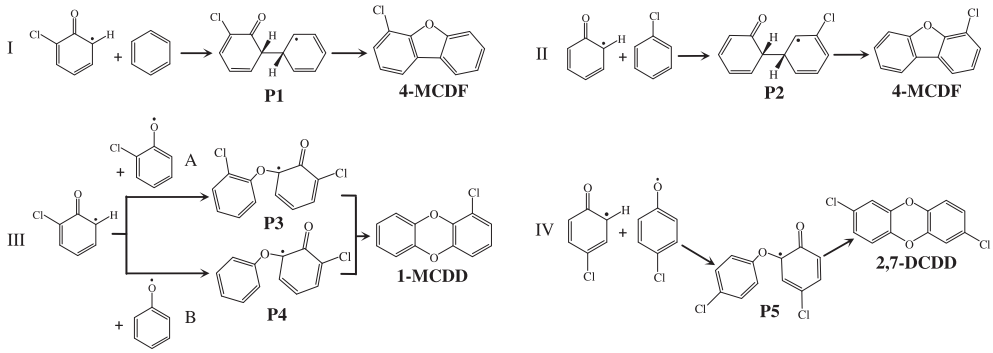

Fig. 5. Postulated pathways for the formation of 4-MCDF, 1-MCDD and 2,7-DCDD.

the resonance forms of the phenoxy radical with the unpaired electron delocalised on the carbon atoms *ortho* to the carbon bonded to oxygen. For instance, 2-chlorophenoxy radical have two types of carbon centred radicals, for carbon attached to either chlorine or hydrogen atoms, as illustrated in Fig. S2. However, in the case of captan, the concentrations of benzene and chlorobenzene are much higher than those of phenoxy radicals. For this reason, we expect that the formation of 4-MCDF proceeds by reactions of benzene and chlorobenzene with phenoxy and chlorophenoxy radicals, as depicted by pathways I and II of Fig. 5. Pathway I involves the coupling of the carbon (hydrogen) centred 2-chlorophenoxy radical to benzene while pathway II proceeds via the coupling of the carbon (hydrogen) centred phenoxy radical to CB, forming the P1 and P2 intermediates respectively. H atoms attached on the pivot bridge of these intermediates are loosely held to C atoms. Our quantum chemical calculations reveal that the H—C bonds of the pivot bridge are longer than the H—C bonds in benzene [18]. These hydrogen atoms can depart P1 and P2 intermediates either by self-ejection or through reactions with radical pool formed in the oxidation of captan. 4-MCDF is then produced via cyclising of the phenoxy O onto the adjacent phenyl ring. Similarly, we elucidate the formation of the other MCDF isomers by the reactions involving 3- and 4-chlorophenoxy radicals. The yields of 2- and 4-chlorophenoxy radicals exceed that of 3-chlorophenoxy radical since the *ortho* and *para* positions of CB are more active than the *meta* positions. Thus, the formation of 4-MCDF and 2-MCDF is more favourable, compared to the other two MCDF isomers.

Pathway III (Fig. 5) depicts the proposed routes to 1-MCDD. The recombination of the oxygen-centred 2-chlorophenoxy radical with the carbon (hydrogen) centred radical (pathway IIIA) has been widely documented to form a keto-ether [13,29]. The abstraction of H by radicals leads to the formation of P3 intermediate. The intra-annular displacement of Cl closes the ring to form 1-MCDD. Pathway IIIB proposes another route that involves the coupling of the oxygen-centred nonchlorinated phenoxy radical to the carbon (hydrogen) centred 2-chlorophenoxy radical, owing to the relatively high concentration of benzene. Similarly, the formation of 2-MCDD may be attributed to the reaction of corresponding phenoxy radicals. A higher yield of 1-MCDD than that of 2-MCDD confirms that the *ortho* position of CB is more active than other locations.

To confirm the identification of DCDD, we injected genuine standards of three DCDD isomers; i.e., 1,3-, 2,7- and 2,3-DCDD (Fig. S4). Figure 4 demonstrates that these isomers dominate the distribution of DCDD (>80%). 2,7-DCDD is the most abundant isomer in our measurements, owing to a higher yield of CB in comparison of those of DCB. The oxidation of chlorobenzene favours the formation of 2-/4-chlorophenoxy radicals. As discussed earlier, the recombination of 2-chlorophenoxy radicals normally results in DD and 1-MCDD [13]. The combination of 4-chlorophenoxy radicals produces exclusively 2,7-DCDD, as illustrated by pathway IV in Fig. 5. Two authentic standards of DCDF were also injected; viz., 2,8-DCDF and 2,3-DCDF (Fig. S5). We found 2,3-DCDF as the most abundant isomer in all measurements, consistently with the quantification of *ortho*-DCB as the most abundant DCB isomer. This indicates the importance of the reaction between 3,4-dichlorophenoxy radical and benzene to from 2,3-DCDF, with the pathway similar to that of the formation of 4-MCDF.

Our standards of TriCDF included two isomers with two Cl attached at the *meta* positions of one aromatic ring and one Cl on the other ring: 2,4,8-TriCDF and 1,3,6-TriCDF (Fig. S6). The amount of the two isomers produced in the experiment only comprises a fraction of the total yield of TriCDF (<10%). An important pathway of the TriCDF formation would lead through a recombination of dichlorophenoxy radicals with CB. The low yield of 2,4,8-TriCDF and 1,3,6-TriCDF is a consequence of small yield of *meta*-

Table 3
Distribution (%) and yield (ng g^{-1}) of each group of TCDF isomers at different RT.

Group	Cl substituted positions	Distribution (*Yield*)		
		0.5 s	1 s	5 s
A	[a]1238, [a]1678, [a]1236, [b]1249, [b]2346, [c]2347, 2348, 1239	51 (*144*)	46 (*96*)	33 (*52*)
B	2467, 2368, [c]1279	22 (*62*)	23 (*48*)	21 (*33*)
C	1278, 1267, 2378, [d]2367, [d]3467, 1289	27 (*76*)	31 (*65*)	46 (*72*)

[a,b,c,d] Tentatively identified due to the co-elution as one peak.

DCB, compared to that of the other two DCB isomers.

TriCDD is the least abundant homologue group detected in all the experiments. The injection of two available standards indicated that 2,3,7-TriCDD comprised around 25% of TriCDD yield while a small amount of 1,2,3-TriCDD was also observed (Fig. S7). With no trichlorobenzene (TriCB) detected in VOC analysis, the identification of 1,2,3-TriCDD suggests either an intramolecular transfer of Cl or further chlorination of dichlorophenoxy radicals during their recombination with unchlorinated phenoxy radical. The identification of 1,2,3-TrCDD is an important finding, as it reveals that further chlorination of DCB [12] or PCDD/F occurs in our system.

The formation of TCDF in the gas-phase condensation of phenoxy radicals has been extensively studied [30], including carbon-carbon coupling of two dichlorophenoxy radicals and trichlorophenoxy/monochlorophenoxy radicals at unchlorinated *ortho* sites. The abstraction of H on the diketo dimer by radicals leads to tautomerisation followed by displacement of hydroxyl to form TCDF. Owing to the higher concentrations of chlorinated benzenes than those of phenoxy radicals, we may expect that the reactions between phenoxy radicals and chlorinated benzenes also play a significant role in the formation of TCDF. The elution order of reference chromatogram obtained on the same VF-5 ms column (60 m × 0.25 mm i.d. × 0.25 μm film) [31] enables us to identify several TCDF isomers (Fig. S8). To compare conveniently, those isomers can be categorised into three groups (Table 3) based on the number of Cl on each ring and the substituted positions: (A) three Cl and one Cl; (B) two *ortho* Cl and two *meta* Cl; (C) two *ortho* Cl and two *ortho* Cl.

Both the distribution and yield of group A TCDF exhibit decreasing trend with longer RT. The absence of TriCB and the presence of group A TCDF congeners again suggest the chlorination of TriCDF or further chlorination of dichlorophenoxy radicals during their coupling to chlorobenzene, leading to TCDF. The longer RT reduces the extent of chlorination. Similarly, the yield of group B TCDF decreases as RT increases in spite of their stable contribution to the total TCDF. The *meta*-DCB only contributes a small fraction to the total yield of DCB. Thus, a further chlorination of chlorophenoxy radicals during their recombination with *ortho*-DCB mainly contributes to the formation pathways of group B, owing to the relatively high yield of CB. Consistently, the dechlorination process during the formation of group B may become important at longer RT. Although the contribution of group C TCDF to the total increases with RT, the yield of group C remains relatively stable. With the most toxic isomer 2,3,7,8-TCDF included in this group, the equivalence (TEQ) corresponds to 0.47–0.76 ng (g captan)$^{-1}$. The formation of isomers in group C TCDF involves the oxidation of *ortho*-DCB to dichlorophenoxy radicals and their combination with *ortho*-DCB.

4. Conclusions

This study has established the pathways of formation of PCDD/F during oxidation of captan, a common fungicide. We have carefully determined yields of both PCDD/F and their VOC precursors, using HRGC-ITMS and HRGC-QMS analyses, respectively, and applied the results of quantum chemical calculations to interpret the experimental measurements. In particular, we have quoted the yields of mono to tri chlorinated DD/F and described their formation pathways. This chlorinated DD/F congeners are seldom reported in the literature, in spite of their importance in unravelling the formation of more chlorinated PCDD/F. The VOC analyses detected no phenol or chlorophenols, and revealed benzene and chlorinated benzenes as the PCDD/F precursors. The oxidation of captan favours the formation of PCDF compared to PCDD, by a factor of 13–20, with the total yield of PCDD/F increasing with the residence time. We have identified 4-MCDF as the most abundant chlorinated DD/F and have elucidated its formation pathways. 2,3-DCDF and 2,7-DCDD were the most plentiful dichlorinated isomers in the DCDD/F homologue groups with their abundance related to the availability of *ortho* and *para*-DCB, since *ortho* and *para* positions are more attractive than *meta* in chlorination of chlorobenzene. Finally, the 2,3,7,8-TCDF, the most toxic PCDD/F congener, was also detected in all experiments. Our findings

have direct application to assess the toxic pollutants emitted from chemical fires and from combustion of biomass contaminated with captan.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.07.069.

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